Plan the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use circuit and motion measurements to plan a motor-car test where you change one thing at a time.

SUCCESS CRITERIA - I can...

1. Draw the battery-switch-motor circuit.
2. Predict effects of an open switch and a changed connection.
3. Plan one controlled performance comparison.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “circuit” and “motor” to explain your answer. Sentence starters are optional.

circuit: a connected arrangement of electrical components

motor: a device converting electrical energy to mechanical output

constraint: a requirement or limit that the design must meet

01 Warm-up: bring back an earlier idea

The same flux change through the same coil occurs in **1/3** of the time. **Compare induced-emf magnitude** and **name one thing** that must stay the same for that comparison.

Word help: flux = a measure of magnetic field passing through a chosen area

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

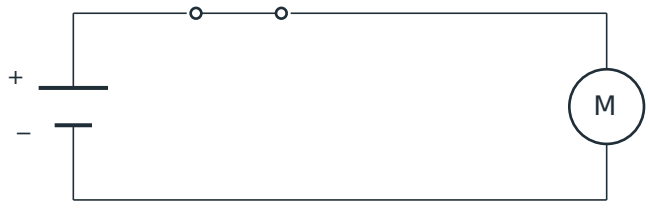
02 Worked example: explain the prediction

Reference: P = IV; E = PΔt.
Treat power as constant. Use approved battery, switch and motor. Disconnect before rewiring; keep wheels clear for first power.

Worked reasoning: A classroom car circuit uses a battery, switch and motor in one closed loop. The switch opens the current path; a chassis drawing must also reserve space for safe wiring and moving parts.

Your turn: Draw the circuit separately from a labeled component-placement plan.

Sentence starter: My design includes ... . It does what is needed by ... .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Draw and **label the battery**, switch, motor and wires as a complete loop. **Mark a location where an** open connection stops current.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: ① ② ③

04 Apply the physics

During a supplied small classroom car test, the motor draws **0.50 A** at **3 V** for **6 s** and the car travels **5 m**. **Find electrical input power**, input energy and average speed. **Complete the table** and keep these supplied values separate from your team’s later measurements.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Voltage (V)	Current (A)	Time (s)	Distance (m)
3	0.5	6	5

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Your team changes wheels, battery and axle alignment together and gets a faster run. Can you attribute the change to wheels alone? **Explain a better comparison**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Electrons in a wire drift at **0.4 mm/s** while a lamp responds quickly after a switch closes. **Explain why response time cannot** be found by timing one electron across the wire.
Draw the stated circuit and **label quantities relevant** to the question.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Name two reasons electrical input energy will exceed the car's final translational kinetic energy.

Sentence starter: I notice … . This supports … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **6 V** motor draws **0.8 A** normally and **2.4 A** when stalled in supplied test data. **Compare input powers** and **explain a test plan** that avoids prolonged stall.

Sentence starter: Both … . They differ in … because … .

Condition	Voltage (V)	Current (A)	Power (W)
Running	6	0.8	____
Stalled	6	2.4	____

Marzano check - circle: 1 2 3